

Properties of matter

Main properties

There are many different properties of matter.



Boiling point is the hottest a liquid can get before becoming a gas.



Freezing point is the temperature at which a liquid becomes a solid.



Plasticity is how well a solid can be reshaped.



Conductivity is how well a material lets electricity or heat travel through it.



Malleability is how well a solid can be shaped without breaking.



Tensile strength is how much a material can stretch without breaking.



Flammability is how easily and quickly a substance will catch fire.



Reflectivity is how well a material reflects light. Water reflects well.



Transparency is how well a material will let light pass through it.



Flexibility is how easily a material can be bent.



Solubility is how well a substance will dissolve, such as salt in water.

Some materials are hard and brittle, while others are flexible. Some materials are colorful, while others are transparent. These kinds of features are called “properties.”



A cork floats on oil.
Oil floats on water.

Does it float?

It's easy to learn about some properties, such as the ability to float. The amount of matter in a certain volume of an object is called its density. Objects and liquids float on liquids of a higher density and sink through liquids of a lower density.

A plastic building brick sinks through oil but floats on water.

An onion sinks through oil and water, but floats on syrup.
Syrup sinks below water.

A good insulator

Heat cannot easily pass through some materials.

These are known as insulators. For example, aerogel can completely block the heat of a flame. But don't try this at home!

